

AQIL YUSRI

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PROFESSIONAL SUMMARY

Physics undergraduate with hands-on experience across full hardware stack: from fabricating and characterizing optical fiber sensors, to designing CMOS circuits in the SkyWater Sky130 PDK, to taking Verilog RTL through a complete RTL-to-GDS flow with timing signoff. Skilled in analog and digital IC design, simulation (NGSpice, XScem), physical design and layout verification (KLayout, Librelane), and functional verification with cocotb. Experienced in PVT corner analysis, device characterization, and quantitative reporting, with embedded and PCB prototyping background (ESP32, Raspberry Pi, KiCad). Seeking an Engineering Internship in IC design, semiconductor, or hardware and electronic roles.

PROJECTS

OTDR-Based D-Shaped Optical Fiber Temperature Sensor | github.com/AQLYSR/otdr-d-shaped-temperature-sensor

- Fabricated D-shaped optical fiber temperature probes using controlled mechanical side-polishing, producing 5 configurations with 2, 4, 6, 7 and 8 dB insertion loss to study the effect of polishing depth on sensing performance.
- Characterized OTDR-based point sensing performance from 40-80 °C in 5°C increments by analyzing backscattered signal attenuation.
- Achieved the best overall sensing performance with 7 dB D-shaped probe, obtaining 0.0008 dB/°C sensitivity and $R^2 = 0.9552$ at 5 ns pulse width.

CMOS Ring Oscillator Design | github.com/AQLYSR/cmos-ring-oscillator

- Designed a 5-stage CMOS ring oscillator in XScem using SkyWater Sky130 (130 nm) PDK, with 1.0 μm $\times$ 0.15 μm CMOS inverter device at 1.8 V supply.
- Characterized the CMOS inverter using DC and transient simulations, extracting a 0.833 V switching threshold and 22.11 ps average propagation delay.
- Achieved 3.99 GHz simulated oscillation frequency at the typical process corner, 1.8 V and 27 °C, within 12 % of the 4.52 GHz analytical prediction.
- Performed PVT sensitivity analysis across 1.6-2.0 V, -10-80 °C, and ss/tt/ff process corners, observing ~96%/V voltage sensitivity and a 1.99 $\times$ frequency spread between slow and fast corners.

Digital PWM Generator Design | github.com/AQLYSR/pwm-generator

- Designed a synthesizable 8-bit PWM generator in Verilog using a counter-comparator architecture, providing 256-step duty-cycle resolution (0-99.6%) with a 390.6 kHz output at 100 MHz clock.
- Developed cocotb functional verification with Icarus Verilog, validating 25% duty-cycle accuracy within $\pm 1\%$ and the 0% duty-cycle boundary case, achieving 2/2 passing tests.
- Implemented complete RTL-to-GDS flow using Librelane and the Sky130A PDK, including synthesis, floorplanning, placement, CTS, routing, and GDS generation.
- Achieved +4.95 ns worst-case setup slack and +0.108 ns hold slack across 9 PVT signoff corners, with 0 timing violations at the 100 MHz target clock.

EXPERIENCE

International Office, Universiti Sains Malaysia

Feb 2025 - Mar 2025

Internship

- Competently coordinated orientation programmes for ~200 incoming international students, ensuring seamless registration, communication, scheduling and on-site support within 2 consecutive months.
- Led an exchange program involving students from Nihon Fukushi University, Japan.
- Supported administrative operations, including emailing incoming international students, assisting officers with passport and visa documentation, photographing orientation week activities, improving administrative efficiency by 40%.

EDUCATION

Universiti Sains Malaysia (USM)

2023 - 2027

Bachelor of Science (Physics) with Honours

- CGPA: 3.69
- Wakalah Paynet-Yayasan Khazanah Scholarship

Universiti Teknologi MARA (UITM)

2022 - 2023

Foundation in Science

- CGPA: 3.89

CERTIFICATIONS

- Semiconductor Devices Courses, Dreamcatcher (2025)
- Advanced Semiconductor Process Technologies, Dreamcatcher (2025)
- Semiconductor Crystal Growth and Substrate Preparation, Dreamcatcher (2025)
- JLPT N3, Japan Foundation (2026)

EXTRACURRICULAR

- Member, USM Buddies — Universiti Sains Malaysia (2023 – 2026)
- Volunteer, Penang International Science Festival (PISF) (2025)

SKILLS

Software & Tools: XScem, NGSpice, Sky130 PDK, Verilog, Icarus Verilog, Librelane, KLayout, KiCad, ESP32, Raspberry Pi, OTDR, Python, C++, Git, Linux, Docker

Technical Skills: CMOS circuit design, DC/transient analysis, PVT corner analysis, static timing analysis, floorplanning, soldering, circuit testing & troubleshooting, device characterization, technical reporting & documentation

LANGUAGES

- Malay:** Native Proficiency
- English:** Full Professional Proficiency
- Japanese:** Limited Proficiency (JLPT N3)